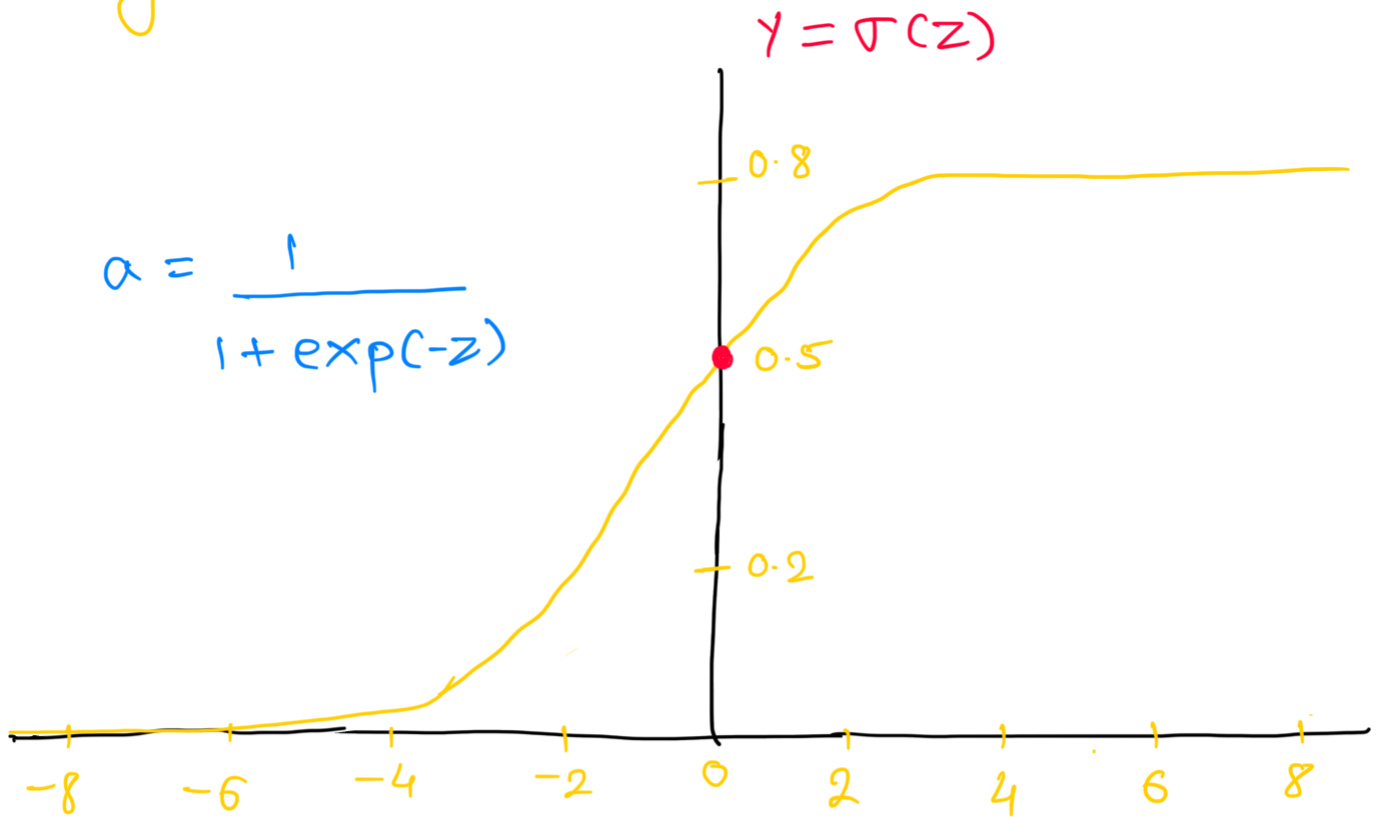


Sigmoid Function



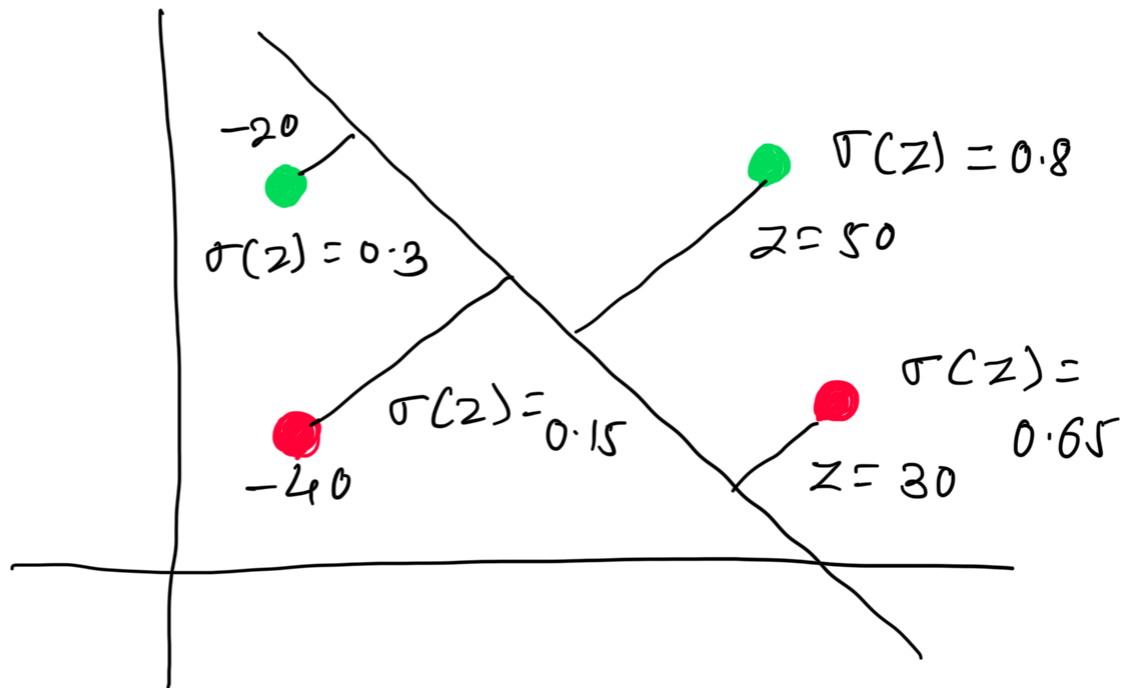
$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Impact of Sigmoid

$$w_n = w_0 + \eta(y_i - \hat{y}_i)x_i$$

$$\hat{y}_i = \sigma(z)$$

Where, $z = \sum w_i x_i$



y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
1	0.8	0.2
0	0.65	-0.65
1	0.3	0.7
0	0.15	-0.15

$$w_n = w_0 + \eta \times 0.2 \times x_i$$

$$w_n = w_0 - \eta \times 0.65 \times x_i$$

$$\omega_n = \omega_0 + \eta \times 0.7 \times \mathcal{R}_i$$

$$\omega_n = \omega_0 + \eta \times 0.15 \times \mathcal{R}_i$$

Magnitude of ω_n depends on distance
between line and point.